Completeness theorem, second version

The Open Logic Project contributors

openlogic:first-order-logic/completeness:local-snapshot-20261013;
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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the closing prov-inconsistency step is a tagrefs block naming the per-proof-system fact and is delivered as the author wrote it

Editorial note. The label cor:completeness names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own cor environment in the unit source.

Allowed standard prerequisites (editorial).

- Completeness theorem, first version. Same unit, \ollabel{thm:completeness} (registered card). Contraposed: not satisfiable implies inconsistent; applied to Gamma union {not A}.
- Entailment to unsatisfiability. Unit olp-first-order-logic-syntax-and-semantics, \ollabel{prop:entails-unsat}. Gamma entails A makes Gamma union {not A} unsatisfiable.
- Inconsistency explodes derivability. If Gamma union {not A} is inconsistent then Gamma proves A (the per-proof-system prov-inconsistency fact, cited through the tagrefs block).. Final step: converts the inconsistency of Gamma union {not A} into the derivation Gamma proves A.

Source reference. Original theorem thm-completeness: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 1003–1003. Prerequisite pointer invoked at proof line(s) 1042; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-entails-unsat: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 2227–2227. Named cross-unit citation: the target label is staged in the corpus under unit olp-first-order-logic-syntax-and-semantics at source line 2227; the cited fact

is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Corollary 0.1 (Completeness Theorem, Second Version)**.** *For all* Γ *and sentences* A*: if* $\Gamma \models A$ *then* $\Gamma \vdash A$*.*

Proof. Note that the Γ's in 0.1 and thm-completeness are universally quantified. To make sure we do not confuse ourselves, let us restate thm-completeness using a different variable: for any set of sentences Δ, if Δ is consistent, it is satisfiable. By contraposition, if Δ is not satisfiable, then Δ is inconsistent. We will use this to prove the corollary.

Suppose that $\Gamma \models A$. Then $\Gamma \cup \{\neg A\}$ is unsatisfiable by prop-entails-unsat. Taking $\Gamma \cup \{\neg A\}$ as our Δ, the previous version of thm-completeness gives us that $\Gamma \cup \{\neg A\}$ is inconsistent. By prfAX/fol:axd:prv:prop:prov-incons, prfSC/fol:seq:prv:prop:prov-incons, prfND/fol:ntd:prv:prop:prov-incons, prfTab/fol:tab:prv:prop:prov-incons, $\Gamma \vdash A$. □